BMO1 2025 (November 2025), Problem 5

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The problem

George defines a sequence of positive integers $t_1, t_2, t_3, \ldots$ as follows. He first sets $t_1 = 1$. Then, for $n \geq 1$:

- *If t_n is even, then $t_{n+1} = t_n/2$.*
- *If t_n is odd and greater than 1, then t_{n+1} is $t_n/3$ rounded to the nearest integer.*
- *If $t_n = 1$, then $t_{n+1} = 2025k$ where k is the number of terms equal to 1 amongst $t_1, t_2, \ldots, t_n$.*

Does this sequence contain every positive integer?

The answer is yes. For odd t the number $t/3$ is never halfway between two integers, so the rounding is never ambiguous. The sequence starts

$$1,\ 2025,\ 675,\ 225,\ 75,\ 25,\ 8,\ 4,\ 2,\ 1,\ 4050,\ 2025,\ 675,\ \ldots$$

We write $a \to b$ if b is the term after a, and call a an *ancestor* of m if the rules for terms greater than 1, applied from a, reach m.

Script A **10/10**

The answer is yes.

Note that if $t_n > 1$ then $1 \leq t_{n+1} < t_n$ (for odd $t_n \geq 3$, the rounded value of $t_n/3$ lies between 1 and $(t_n+1)/3 < t_n$). So after each 1 the sequence decreases back to 1, hence 1 appears infinitely often and the term after the kth 1 is $2025k$. If $2025k$ is an ancestor of $m \geq 2$, the terms after $2025k$ decrease and stay above 1 until m is reached, so m appears. Hence we only need every $m \geq 2$ to have an ancestor divisible by $2025 = 3^4 \cdot 5^2$.

Tracing backwards, $2m \to m$ for every m. If m is odd then $3m$ is odd, so $3m \to m$. If m is even then $3m \pm 1$ is odd and $(3m \pm 1)/3 = m \pm \frac{1}{3}$ rounds to m, so $3m \pm 1 \to m$.

Step 1 (an odd ancestor n with $5 \nmid n$). *If m is even, $3m - 1$ and $3m + 1$ differ by 2, so one of them is not a multiple of 5, and n is that one. If m is odd and $5 \nmid m$, take $n = m$. If m is odd and $5 \mid m$, take $n = 6m + 1$, since $6m + 1 \to 2m \to m$ and $n \equiv 1 \pmod 5$.*

Step 2 (an odd ancestor divisible by 25). *Modulo 25, the powers $2^1, 2^2, \ldots, 2^{20}$ are*

$$2,\ 4,\ 8,\ 16,\ 7,\ 14,\ 3,\ 6,\ 12,\ 24,\ 23,\ 21,\ 17,\ 9,\ 18,\ 11,\ 22,\ 19,\ 13,\ 1,$$

which are all twenty residues not divisible by 5. As $5 \nmid 3n$, there is $1 \leq j \leq 20$ with $3 \cdot 2^j n \equiv 1 \pmod{25}$. Let $e = 2^j n$, so halving j times takes e to n. Then $a = 3e - 1$ is odd, $a \to e$, and $25 \mid a$.

Step 3 (four factors of 3). *Since a is odd, $81a \to 27a \to 9a \to 3a \to a$, and $2025 \mid 81a$.*

Hence every m has an ancestor divisible by 2025, and every positive integer appears.

Marker's comment. Complete. The reduction to an ancestor divisible by 2025 is justified, the backward steps carry the right parity conditions (a common error in the markers' report), and the residues of 2^j modulo 25 are listed rather than claimed, with the odd multiples of 5 dealt with first.

Script B 6/10

Every term greater than 1 *is followed by a smaller positive integer, so the sequence keeps returning to* 1*, and after the kth* 1 *comes* $2025k$*. So the problem reduces to showing every* m *can be reached from some multiple of* 2025 *by halving and dividing by* 3*.*

Working backwards, $2m \to m$ *always,* $3m \to m$ *when* m *is odd, and* $3m \pm 1 \to m$ *when* m *is even. If* m *is even replace* m *by whichever of* $3m \pm 1$ *is not a multiple of* 5*, so we may assume* m *is odd.*

The powers of 2 *modulo* 25 *are* $2, 4, 8, 16, 7, 14, 3, 6, 12, 24, 23, 21, 17, 9, 18, 11, 22, 19, 13, 1$*, so they cover every residue coprime to* 5*, and there is* $j \geq 1$ *with* $3 \cdot 2^j m \equiv 1 \pmod{25}$*. Then* $a = 3 \cdot 2^j m - 1$ *is odd,* $a \to 2^j m \to \cdots \to m$*, and* $25 \mid a$*. Finally* $81a \to 27a \to 9a \to 3a \to a$ *and* $2025 \mid 81a$*, so* m *appears.*

Marker's comment. Nearly all of the proof, with the parity conditions and the residue table done properly. If m is odd and $5 \mid m$, then $3 \cdot 2^j m$ is never 1 modulo 25, so the argument fails for $m = 5, 15, 25, \ldots$. The extra step $6m + 1 \to 2m \to m$ repairs the gap.

Script C 3/10

The sequence splits into blocks, each starting at $2025k$ *and falling to* 1*, since every term greater than* 1 *is followed by a smaller one. So* $2025k$ *appears for every* k*, and* m *appears if some multiple of* 2025 *leads down to* m*.*

Backwards, $2m \to m$*, and* $3m \to m$ *for odd* m*, and* $3m \pm 1 \to m$ *for even* m*. Take* m *odd and not a multiple of* 5 *(otherwise first use* $3m \pm 1$ *or* $6m + 1$ *as needed). Since* 2 *and* 25 *have no common factor, the powers of* 2 *take every value modulo* 25 *that is coprime to* 5*. So some* j *gives* $3 \cdot 2^j m \equiv 1 \pmod{25}$*, and* $a = 3 \cdot 2^j m - 1$ *is an odd multiple of* 25 *with* $a \to 2^j m \to \cdots \to m$*. Then* $81a$ *is a multiple of* 2025 *leading to* m*.*

Marker's comment. The structure is right, but the residue step rests on a false principle. Being coprime to 25 does not make a number's powers cover every residue coprime to 5, since the powers of 11 give only 11, 21, 6, 16, 1. The claim for 2 is true but needs the twenty residues checked, and the report gave this argument no marks, so the credit is for the blocks and the backward steps.

Script D 1/10

Each time the sequence reaches 1 *the next term is* $2025k$*, and then the terms decrease back to* 1*. So the sequence is made of blocks from* $2025k$ *down to* 1*.*

If $t = 3x + 1$ *then* $t/3 = x + \frac{1}{3}$*, which rounds to* x*, so the term after* $3x + 1$ *is* x*. Also the term after* $2x$ *is* x*. So going backwards from any* x *we can reach* $3x + 1$*,* $2x$*,* $3(3x + 1) + 1$ *and so on, and these numbers cover every residue class modulo* 2025*. So some multiple of* 2025 *leads down to* x*, and the sequence contains every positive integer.*

Marker's comment. The block structure earns the mark. The step $3x + 1 \to x$ needs x even (for $x = 3$ the term after 10 is 5), and the claim that the backward numbers cover every residue modulo 2025 is the whole problem, asserted without argument.

What full marks needs

Show that every m has an ancestor divisible by 2025, with the right parity conditions on each backward step. The factor 25 comes from doubling, once the residues of 2^j modulo 25 are checked and odd multiples of 5 removed, and the factor 81 from tripling an odd number four times.